

HAIRPINS AND BASEPAIRS IN RNA SECONDARY STRUCTURES ARE ASYMPTOTICALLY JOINTLY NORMAL: PROOF OF A CONJECTURE OF BU, KAUERS AND ZEILBERGER

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ABSTRACT. Bu, Kauers and Zeilberger conjectured that the numbers of hairpins and of basepairs in a uniformly random RNA secondary structure on n vertices are, after centering and scaling, asymptotically bivariate normal with correlation $\frac{1}{2}\sqrt{5\sqrt{5}-11} = 0.21233\dots$. We give a rigorous proof. The method is the standard one (a square-root singularity that moves analytically with the marking variables, hence a quasi-power expansion of the moment generating function); a bivariate normal law for these two parameters was derived along these lines, at a heuristic level and for a family of models containing this one, by Cuesta and Manrubia in 2017, and the statement also follows from a general theorem of Drmota once its hypotheses are checked. Our contribution is to check them: we give a short self-contained proof with a uniform coefficient estimate, compute the limiting mean vector and covariance matrix exactly, and verify every algebraic identity used both in Mathematica and in the Lean 4 proof assistant.

1. STATEMENT

A *secondary structure* on $[n] = \{1, \dots, n\}$ is a set of pairwise disjoint pairs $\{i, j\}$ (*basepairs*) with $j - i \geq 2$ and no crossings (no $i < k < j < l$ with $\{i, j\}, \{k, l\}$ both present). A basepair $\{i, j\}$ is a *hairpin* if none of $i + 1, \dots, j - 1$ is paired. For a uniformly random secondary structure on $[n]$ let X_n be its number of hairpins and Z_n its number of basepairs. In [3] the following is conjectured, with a pledge of \$100 to the OEIS Foundation for a proof.

Theorem 1. *As $n \rightarrow \infty$,*

$$\left(\frac{X_n - \mathbb{E}X_n}{\sqrt{\text{Var } X_n}}, \frac{Z_n - \mathbb{E}Z_n}{\sqrt{\text{Var } Z_n}} \right) \xrightarrow{d} \mathcal{N}\left(0, \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}\right), \quad c = \frac{\sqrt{5\sqrt{5}-11}}{2}.$$

Moreover, with

$$\mu = \left(1 - \frac{2}{\sqrt{5}}, \frac{5-\sqrt{5}}{10}\right), \quad H = \begin{pmatrix} 2 - \frac{22}{5\sqrt{5}} & \frac{25-11\sqrt{5}}{50} \\ \frac{25-11\sqrt{5}}{50} & \frac{1}{10\sqrt{5}} \end{pmatrix}, \quad \det H = \frac{13\sqrt{5}-29}{50} > 0,$$

we have $(\mathbb{E}X_n, \mathbb{E}Z_n) = n\mu + O(1)$ and the covariance matrix of (X_n, Z_n) is $nH + O(1)$.

Remark 2. The density displayed in [3], $\frac{\sqrt{1-c^2}}{2\pi} e^{-x^2/2-y^2/2+cxy}$, is that of the centered normal law with covariance matrix $(1-c^2)^{-1} \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}$, whose marginals have variance $1/(1-c^2) = 1.0472\dots$; this cannot be the limit of coordinates standardized to variance one. We read the conjecture as its text and abstract state it: bivariate normal with unit variances and correlation c , whose density is $\frac{1}{2\pi\sqrt{1-c^2}} \exp\left(-\frac{x^2-2cxy+y^2}{2(1-c^2)}\right)$.

Related work and what is new. Nothing in the method is new. For a quadratic generating function with a simple dominant branch point, Gaussian limit laws follow from the “algebraic singularity schema” [6, Thm. IX.12] (one parameter) and its multivariate versions [1, 7], [5, Thm. 2.25], [8]; see also [6, §IX.12]. For RNA secondary structures specifically:

- Jin and Reidys [10] proved central and local limit theorems for the number of basepairs alone, with the same mean and variance constants $\frac{5-\sqrt{5}}{10}$ and $\frac{1}{10\sqrt{5}}$ (numerically, 0.27639 and 0.04472).
- Cuesta and Manrubia [4, §3.3] treat the pair (basepairs, hairpins) in the family of models with stems of length $\geq s$ and hairpin loops of size $\geq m$. They obtain the expansion (3) below from Darboux’s theorem and read off a bivariate normal law, with numerical constants for $(s, m) = (2, 3)$. Their argument is at the level of rigour customary in physics: the coefficient asymptotics are applied for each fixed value of the marking variables, without the uniformity that the continuity theorem requires, and non-degeneracy is not discussed. The model of [3] is the case $s = m = 1$: their discriminant $\Delta(z, w, u)$ satisfies $(1 - z)^2 \Delta = R$ identically, with R as in Lemma 3. As a check on both computations, our code reproduces all the constants they publish for $(2, 3)$ (file `checks/cuesta_manrubia.wls`).
- Poznanović and Heitsch [12] prove univariate Gaussian laws for motif counts in a stochastic context-free grammar model of RNA folding, also via [6, Thm. IX.12].

What this note adds is a complete proof for the model and normalization of [3]: the hypotheses (a unique simple dominant root, non-vanishing of the square-root coefficient, non-singularity of the Hessian) are verified exactly, the uniform estimate is proved directly (Lemma 5) so that the argument can be read without the general theory, and the constants are obtained in closed form in $\mathbb{Q}(\sqrt{5})$. A reader who prefers to quote a theorem can stop after Section 3: the representation (2), with ρ, A, B analytic for $|X| < 0.51$ and $B(\rho) \neq 0$, supplies the hypotheses of [5, Thm. 2.25] with exponent $\frac{1}{2}$, and the limit is non-degenerate because $\det H \neq 0$ (Section 5).

Disclosure of method. The mathematics in this note was produced by a large language model, Anthropic’s Claude (Claude Fable 5.1, run as the Claude Code agent), working at the author’s direction. The model proposed the proof strategy, wrote the proof and this text, wrote and ran the verification code (Mathematica and Lean, on the author’s computer), and carried out the literature search. A second, independent instance of the same model, given only the manuscript and the code, was asked to referee the proof adversarially; it found no errors, and the gaps in exposition that it identified have been repaired. A third instance checked the citations against the sources; that search is what turned up [4] and [10], which the first draft of this note had missed. The author is not a professional mathematician. His contribution was to select the problem, direct and question the work, and study the argument until he understood its structure; he has not independently verified every estimate. He takes responsibility for the note and will correct any error brought to his attention. We state this at the outset so that the reader can weigh the claims accordingly; the claims that rest on exact computation are identified as such and the code is available at the repository listed below.

Following the practice of [11], we distinguish throughout between what is *proved* (a complete argument is given) and what is *machine-verified* (an identity confirmed by exact symbolic computation in $\mathbb{Q}(\sqrt{5})$ or over $\mathbb{Q}(x, z)$; files `checks/verify.wls`, `checks/grammar_check.wls`). The algebraic identities are in addition checked by the kernel of the Lean 4 proof assistant; see Section 7. Numerical values are quoted only as sanity checks and no step of the proof depends on them.

2. THE GENERATING FUNCTION

Let $f_n(x, z) = \sum_s x^{\text{hairpins}(s)} z^{\text{basepairs}(s)}$ over secondary structures s on $[n]$, and $S(X) = \sum_{n \geq 0} f_n X^n$.

Lemma 3. $S = 1 + XS + zX^2S(S - 1 + (x - 1)\frac{X}{1-X})$. Consequently $S = 1 + X + F$, where

$$F = \frac{-2X^4z + X^3xz + X^2z - X^2 + 2X - 1 + \sqrt{R}}{2X^2z(X - 1)},$$

$$R = X^6x^2z^2 - 2X^5xz^2 - 2X^5xz + 4X^4xz + X^4z^2 - 2X^4z - 2X^3xz \\ + X^4 + 4X^3z - 4X^3 - 2X^2z + 6X^2 - 4X + 1,$$

and $\sqrt{R}$ denotes the power series in X with constant term 1 whose square is R .

Proof. A structure is empty, or vertex 1 is unpaired (factor XS), or 1 is paired to some j : then the structure is the pair (weight zX^2), a structure on the *nonempty* vertex set $\{2, \dots, j-1\}$, and an arbitrary structure on $\{j+1, \dots, n\}$. The inner structure contributes $S-1$, except that when it has no basepair (generating function $X/(1-X)$) the outer pair is a hairpin and gets an extra factor x . This is the functional equation; it determines the power series S uniquely from $S(0) = 1$. That $1 + X + F$ satisfies it is a direct rational-function identity after substituting and using $(\sqrt{R})^2 = R$ (file `checks/grammar_check.wls`). Finally $\sqrt{R} = 1 - 2X + (1-z)X^2 + O(X^3)$, and in fact the numerator of F is $-2zX^4 + O(X^5)$, so $F = X^2 + O(X^3)$ and $1 + X + F$ is a power series with constant term 1. $\square$

This is Theorem 2 of [3] (there proved by symbolic computation [2]), up to the two initial terms $1 + X$, which the formula for F omits. We also checked $[X^n]F = f_n$ for $2 \leq n \leq 14$ against exhaustive enumeration (file `checks/verify.wls`).

3. LOCAL STRUCTURE AT THE DOMINANT SINGULARITY

At $x = z = 1$,

$$(1) \quad R(X, 1, 1) = (X - 1)^2(X^2 - 3X + 1)(X^2 + X + 1),$$

with roots $\rho_0 = \frac{3-\sqrt{5}}{2} = 0.38196\dots$ (simple), 1 (double), $e^{\pm 2\pi i/3}$ and $1/\rho_0$.

Lemma 4. *There are a bounded open neighbourhood $V \subset \mathbb{C}^2$ of $(1, 1)$, on which $|z| \geq \frac{1}{2}$, and an analytic $\rho : V \rightarrow \mathbb{C}$ with $\rho(1, 1) = \rho_0$ and $0 < |\rho| \leq 0.45$ on V , such that for $(x, z) \in V$*

$$R(X, x, z) = \left(1 - \frac{X}{\rho(x, z)}\right)Q(X; x, z),$$

where Q is a polynomial in X whose coefficients are analytic on V , $Q(0; x, z) = 1$, and $Q(X; x, z) \neq 0$ for $|X| \leq 0.51$.

Proof. By (1), $R(\cdot, 1, 1)$ has exactly one zero in $|X| \leq 0.51$, namely the simple zero ρ_0 , and none on $|X| = 0.51$. The coefficients of R are polynomials in (x, z) , so $R(\cdot, x, z) \rightarrow R(\cdot, 1, 1)$ uniformly on the compact circle $|X| = 0.51$, and by Rouché's theorem the same holds for (x, z) near $(1, 1)$; call the zero $\rho(x, z)$. Since $\partial_X R(\rho_0, 1, 1) \neq 0$ it is analytic by the implicit function theorem, and $\rho \neq 0$ because $R(0, x, z) = 1$. Shrink V so that $|\rho| \leq 0.45$ and $|z| \geq \frac{1}{2}$. Dividing the polynomial R by $1 - X/\rho$ gives Q , with coefficients polynomial in $1/\rho$ and the coefficients of R ; its zeros are the remaining zeros of R , all outside $|X| \leq 0.51$; and $Q(0) = R(0) = 1$. $\square$

On the disc $\Delta = \{|X| < 0.51\}$ the function Q is zero-free, so $\sqrt{Q} := \exp(\frac{1}{2} \int_0^X \partial_X Q/Q)$ (integral along the segment; the integrand is jointly analytic) is analytic on $\Delta \times V$, with $\sqrt{Q}(0) = 1$ and $|\sqrt{Q}| = |Q|^{1/2}$. Put $\Phi = X^2 F$ and

$$A = \frac{-2X^4 z + X^3 x z + X^2 z - X^2 + 2X - 1}{2z(X-1)}, \quad B = \frac{\sqrt{Q}}{2z(X-1)},$$

both analytic on $\Delta \times V$, since $|X-1| \geq 0.49$ and $|z| \geq \frac{1}{2}$ there. Since $\sqrt{Q} \cdot (1-X/\rho)^{1/2}$, with the binomial series for the second factor, is a power series with constant term 1 and square R , it equals $\sqrt{R}$. Hence, as power series in X ,

$$(2) \quad \Phi = A + B(1-X/\rho)^{1/2}, \quad f_n = [X^{n+2}] \Phi \quad (n \geq 2).$$

Let $\beta(x, z) = B(\rho(x, z); x, z)$, analytic on V . At $(1, 1)$, $Q(\rho_0) = -\rho_0 \partial_X R(\rho_0, 1, 1) = 36\sqrt{5} - 80 > 0$, so

$$\beta(1, 1) = \frac{\sqrt{36\sqrt{5} - 80}}{2(\rho_0 - 1)} \neq 0,$$

We now replace V by a smaller neighbourhood of $(1, 1)$ whose closure is a compact subset of the original one and on which $|\beta| \geq \beta_{\min} > 0$. Then A and B are bounded on $\{|X| \leq \frac{1}{2}\} \times V$. This V is fixed from now on.

4. A UNIFORM COEFFICIENT ESTIMATE

Lemma 5. *Uniformly for $(x, z) \in V$, as $n \rightarrow \infty$,*

$$f_n(x, z) = b(x, z) \rho(x, z)^{-n} n^{-3/2} (1 + \varepsilon_n(x, z)), \quad b = -\frac{\beta}{2\sqrt{\pi} \rho^2}, \quad \sup_V |\varepsilon_n| = O(1/n).$$

Proof. Write $N = n + 2$. Let M bound $|A|, |B|$ on $\{|X| \leq \frac{1}{2}\} \times V$. By Cauchy's inequality $|[X^k]A| \leq M2^k$. Put $B_1 = (B - \beta)/(X - \rho)$; it is analytic in $|X| \leq \frac{1}{2}$, and by the maximum principle $|B_1| \leq 2M/(0.5 - 0.45) = 40M$, so $|[X^k]B_1| \leq 40M2^k$. From (2),

$$\Phi = A + \beta(1 - X/\rho)^{1/2} - \rho B_1(1 - X/\rho)^{3/2}.$$

Now $[X^m](1 - X/\rho)^\alpha = \rho^{-m} \binom{m-\alpha-1}{m}$, and $\binom{N-3/2}{N} = \frac{\Gamma(N-1/2)}{\Gamma(-1/2)\Gamma(N+1)} = -\frac{1}{2\sqrt{\pi}} N^{-3/2}(1 + O(1/N))$, while $|\binom{m-5/2}{m}| \leq C(m+1)^{-5/2}$; the constant C and the $O(1/N)$ are absolute, and ρ (complex in general) enters only through the exact factor ρ^{-m} . Since $2|\rho| \leq 0.9$,

$$|[X^N] \rho B_1 (1 - X/\rho)^{3/2}| \leq 40MC|\rho|^{1-N} \sum_{k=0}^N (0.9)^k (N-k+1)^{-5/2} = O(|\rho|^{-N} N^{-5/2}),$$

(split the sum at $k = N/2$), and $|[X^N]A| \leq M|\rho|^{-N}(0.9)^N$. Therefore

$$[X^N]\Phi = -\frac{\beta}{2\sqrt{\pi}} \rho^{-N} N^{-3/2} (1 + O(1/N))$$

uniformly on V : relative to the main term, whose modulus is at least $\frac{\beta_{\min}}{2\sqrt{\pi}} |\rho|^{-N} N^{-3/2} (1 + O(1/N))$, the B_1 -term is $O(1/N)$ and the A -term is exponentially small. Finally $N^{-3/2} = n^{-3/2}(1 + O(1/n))$. $\square$

For $x = z = 1$ this gives $f_n \sim b(1, 1) \rho_0^{-n} n^{-3/2}$ with $b(1, 1) = 1.10437 \dots > 0$; numerically $f_n / (b(1, 1) \rho_0^{-n} n^{-3/2}) = 0.99398$ at $n = 400$. Note also that $\varepsilon_n = f_n n^{3/2} \rho^n / b - 1$ is analytic on V , since f_n is a polynomial and b is zero-free.

5. PROOF OF THEOREM 1

Fix $\tau > 0$ with $(e^s, e^t) \in V$ whenever $|s|, |t| < \tau$, and let $P = \{|s|, |t| < \tau\} \subset \mathbb{C}^2$. For $n \geq 2$ the moment generating function is $M_n(s, t) = \mathbb{E}e^{sX_n + tZ_n} = f_n(e^s, e^t)/f_n(1, 1)$, an entire function. By Lemma 5,

$$(3) \quad M_n(s, t) = \exp(nU(s, t) + W(s, t)) \frac{1 + \varepsilon_n(e^s, e^t)}{1 + \varepsilon_n(1, 1)},$$

where $U = -\log(\rho(e^s, e^t)/\rho_0)$ and $W = \log(b(e^s, e^t)/b(1, 1))$ are analytic on P (shrinking τ so that both quotients stay within distance $\frac{1}{2}$ of 1), with $U(0, 0) = W(0, 0) = 0$.

The Taylor coefficients of U . Differentiating $R(\rho(e^s, e^t), e^s, e^t) = 0$ twice at $(0, 0)$ and solving the resulting linear equations in $\mathbb{Q}(\sqrt{5})$ (file `checks/verify.wls`) gives

$$U(s, t) = \mu_1 s + \mu_2 t + \frac{1}{2}(H_{11}s^2 + 2H_{12}st + H_{22}t^2) + O(|s|^3 + |t|^3),$$

with μ and H exactly as in Theorem 1. Since $H_{11} > 0$ and $\det H > 0$, H is positive definite, and $H_{12}/\sqrt{H_{11}H_{22}} = \frac{1}{2}\sqrt{5\sqrt{5}-11} = c$.

Central limit theorem. Fix $\eta = (\eta_1, \eta_2) \in \mathbb{R}^2$ and let $Y_n = n^{-1/2}(X_n - \mu_1 n, Z_n - \mu_2 n)$. For n large, $(s, t) = i\eta/\sqrt{n} \in P$, and by (3),

$$\mathbb{E}e^{i\eta Y_n} = e^{-i\sqrt{n}\mu\eta} M_n(i\eta/\sqrt{n}) = \exp\left(-\frac{1}{2}\eta^\top H\eta + O(n^{-1/2})\right)(1 + O(1/n)) \longrightarrow e^{-\frac{1}{2}\eta^\top H\eta}.$$

By Lévy's continuity theorem $Y_n \xrightarrow{d} \mathcal{N}(0, H)$.

Means and variances. For $n \geq n_0$ we have $\sup_V |\varepsilon_n| \leq \frac{1}{2}$. With Log the principal branch put

$$g_n(s, t) = \text{Log}(1 + \varepsilon_n(e^s, e^t)) - \text{Log}(1 + \varepsilon_n(1, 1)), \quad L_n = nU + W + g_n.$$

Then g_n is analytic on P with $\sup_P |g_n| = O(1/n)$, and by (3) L_n is an analytic logarithm of M_n on P (so $M_n \neq 0$ there) with $L_n(0, 0) = 0$. By Cauchy's estimates all partial derivatives of g_n of order ≤ 2 at the origin are $O(1/n)$. The first and second derivatives at the origin of the logarithm of a moment generating function are the means and the (co)variances, so

$$(\mathbb{E}X_n, \mathbb{E}Z_n) = n\mu + \nabla W(0) + O(1/n), \quad \Sigma_n = nH + \nabla^2 W(0) + O(1/n),$$

where Σ_n is the covariance matrix of (X_n, Z_n) . This agrees with Theorem 3 of [3], including the second-order terms of the means: an exact computation (file `checks/verify.wls`) gives $\nabla W(0) = \left(\frac{13-3\sqrt{5}}{20}, -\frac{3+2\sqrt{5}}{20}\right)$, which equals $(\mu_1(\frac{7}{4} + \frac{11}{20}\sqrt{5}), \mu_2(-\frac{5}{8} - \frac{13}{40}\sqrt{5}))$ as found there.

Standardization. Consequently, for n large (so that the variances are positive),

$$\left(\frac{X_n - \mathbb{E}X_n}{\sqrt{\text{Var } X_n}}, \frac{Z_n - \mathbb{E}Z_n}{\sqrt{\text{Var } Z_n}}\right) = D_n Y_n + d_n,$$

where $D_n \rightarrow \Lambda := \text{diag}(H_{11}^{-1/2}, H_{22}^{-1/2})$ and $d_n = O(n^{-1/2})$, and by Slutsky's lemma the left side converges in distribution to $\mathcal{N}(0, \Lambda H \Lambda)$, i.e. unit variances and correlation c . $\square$

Remark 6. The expansion (3) is the hypothesis of the quasi-power theorems [9], [5, Thm. 2.22], [8]; the last of these also gives a Berry–Esseen bound of order $n^{-1/2}$ for the convergence $Y_n \rightarrow \mathcal{N}(0, H)$.

6. ILLUSTRATION

Figure 1 compares the exact joint distribution at $n = 400$ with the limit law, using the mean $n\mu + \nabla W(0)$ and covariance nH . The total variation distance between the exact law and the lattice Gaussian with the same mean and covariance is 0.0238, 0.0159, 0.0112, 0.0078, 0.0055 for $n = 50, 100, 200, 400, 800$, consistent with a rate $n^{-1/2}$. This is an illustration only (floating-point; file `figure/joint_distribution.py`); nothing in the proof depends on it.

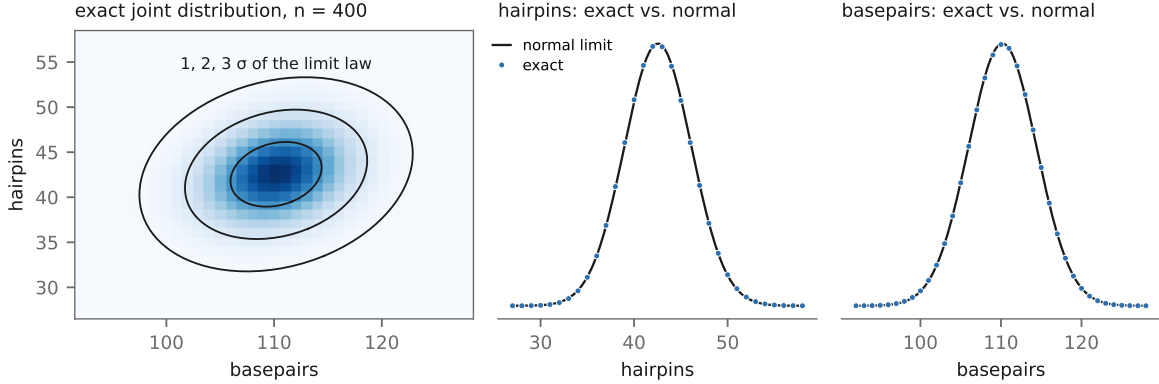


FIGURE 1. Exact joint distribution of (hairpins, basepairs) at $n = 400$ (shading and dots), against the limiting normal law (ellipses and curves).

7. FORMAL VERIFICATION OF THE ALGEBRA

The file `lean_check/LeanCheck/Basic.lean` proves the following in Lean 4 with Mathlib, over an arbitrary field of characteristic zero or, where $\sqrt{5}$ is involved, over $\mathbb{R}$ with a symbol s subject only to $s^2 = 5$ (and $s > 0$ for the inequalities): the factorization (1); the functional equation of Lemma 3 for $S = 1 + X + F$, for any r with $r^2 = R$; $R(\rho_0, 1, 1) = 0$; the complete Taylor expansion of R at $(\rho_0, 1, 1)$, which certifies all partial derivatives of R there, in particular $\partial_X R = 30 - 14\sqrt{5} \neq 0$; the first- and second-order implicit-differentiation identities satisfied by the Taylor coefficients of $\rho(e^s, e^t)$; the values of μ and H obtained from them; $\det H = (13\sqrt{5} - 29)/50 > 0$, $H_{11} > 0$, $H_{12} > 0$, $H_{12}^2 = \frac{5\sqrt{5}-11}{4} H_{11} H_{22}$; and $Q(\rho_0) = 36\sqrt{5} - 80 > 0$. All nineteen theorems are accepted by the kernel with no axioms beyond the three standard ones (`propext`, `Classical.choice`, `Quot.sound`). Mathematica is used only to propose the polynomials; Lean re-verifies them from scratch.

Not formalized: the analytic part of the argument (Lemma 4, Lemma 5, Lévy's theorem) and the elementary calculus that connects the identities above to the derivatives of $U = -\log \rho$. The value of $\nabla W(0)$ is machine-verified in Mathematica only.

Code. All scripts, their outputs and the Lean project are at <https://github.com/bshepp/rna-joint-normality>.

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